

Intelligent stock trading system by turning point confirming and probabilistic reasoning

Depei Bao, Zehong Yang

State Key Lab of Intelligent Technology and Systems,
Computer Science Department, Tsinghua University,
Beijing, China

Expert Systems with Applications, Vol34(2008), PP. 620-627

Presenter: Hung-Hsi Chen
Date: Apr. 22, 2009

Abstract(1/2)

Financial engineering such as trading decision is an emerging research area and also has great commercial potentials. A successful stock buying/selling generally occurs near price trend turning point. Traditional technical analysis relies on some statistics(i.e. Technical indicators) to predict turning point of the trend. However, these indicators can not guarantee the accuracy of prediction in chaotic domain. In this paper, we propose an intelligent financial trading system through a new approach: learning trading strategy by probabilistic model from high-level representation of time series – **turning points** and **technical indicators**. The main contributions of this paper are two-fold.

Abstract(2/2)

First, we utilize high-level representation(turning point and technical indicators). High-level representation has several **advantages** such as insensitive to noise and intuitive to human beings. However, it is rarely used in past research. Technical indicator is the knowledge from professional investors, which can generally characterize the market. **Second**, by combining high-level representation with probabilistic model, the randomness and uncertainty of chaotic system is further reduced. In this way, we achieve great results(comprehensive experiments on S&P500 components) in a chaotic domain in which the prediction is thought impossible in the past.

Architecture of the system

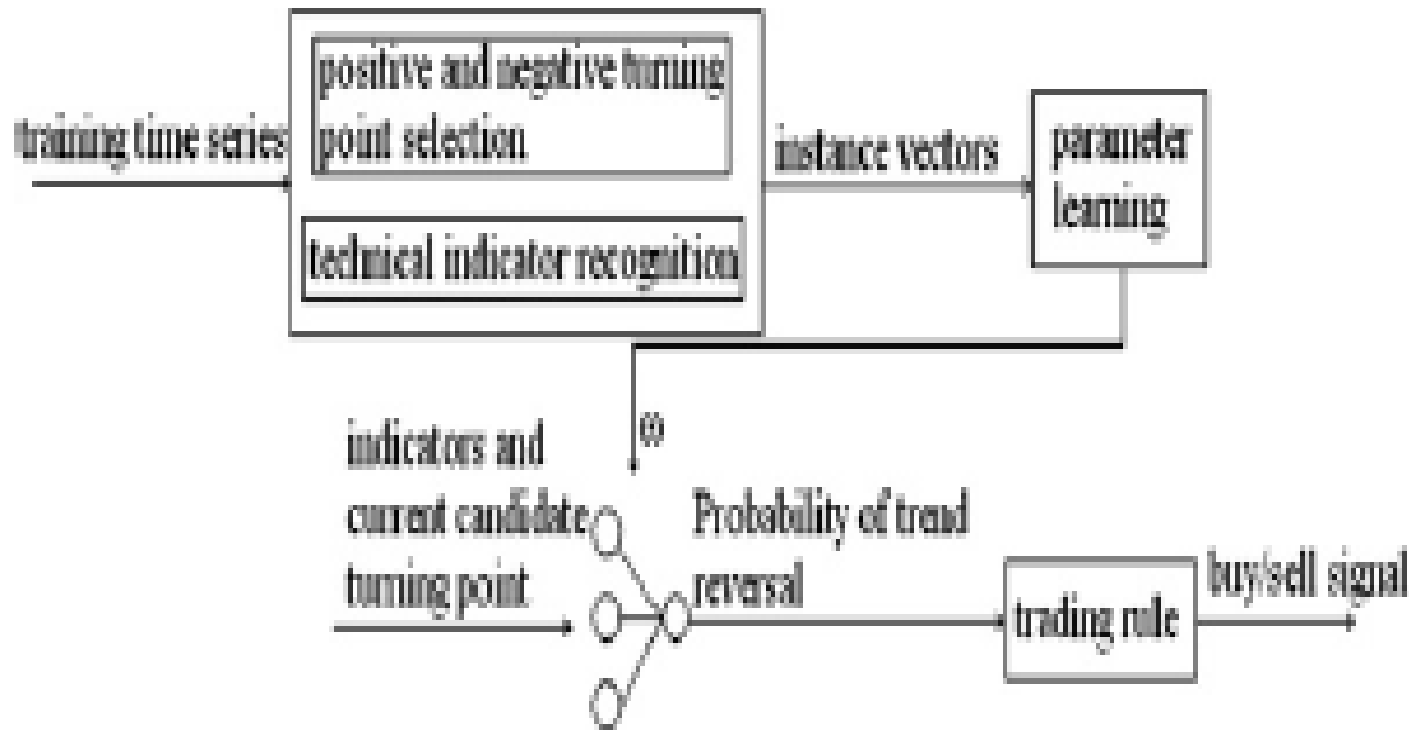


Fig. 1. Architecture of the trading system.

Probabilistic structure

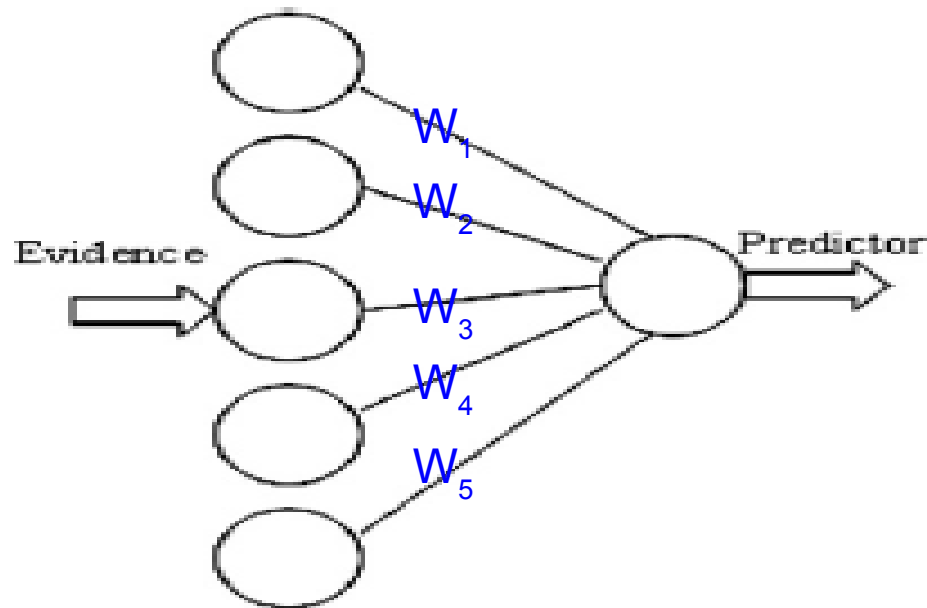


Fig. 2. Probabilistic structure.

- **Every evidence node** represents a technical indicator
- **Predictor** represents whether current candidate turning point is the real trend reversal point(true/false)
- **Parameter** $w = \{w_1, w_2 \dots w_n\}$ is assigned to every link between predictor and evidence.

4 technical indicators are used :

- Moving average system
- RSI oscillator system
- Stochastic slowK-slowD crossover/.oscillator system
- Trendline system

Measure criteria of turning points

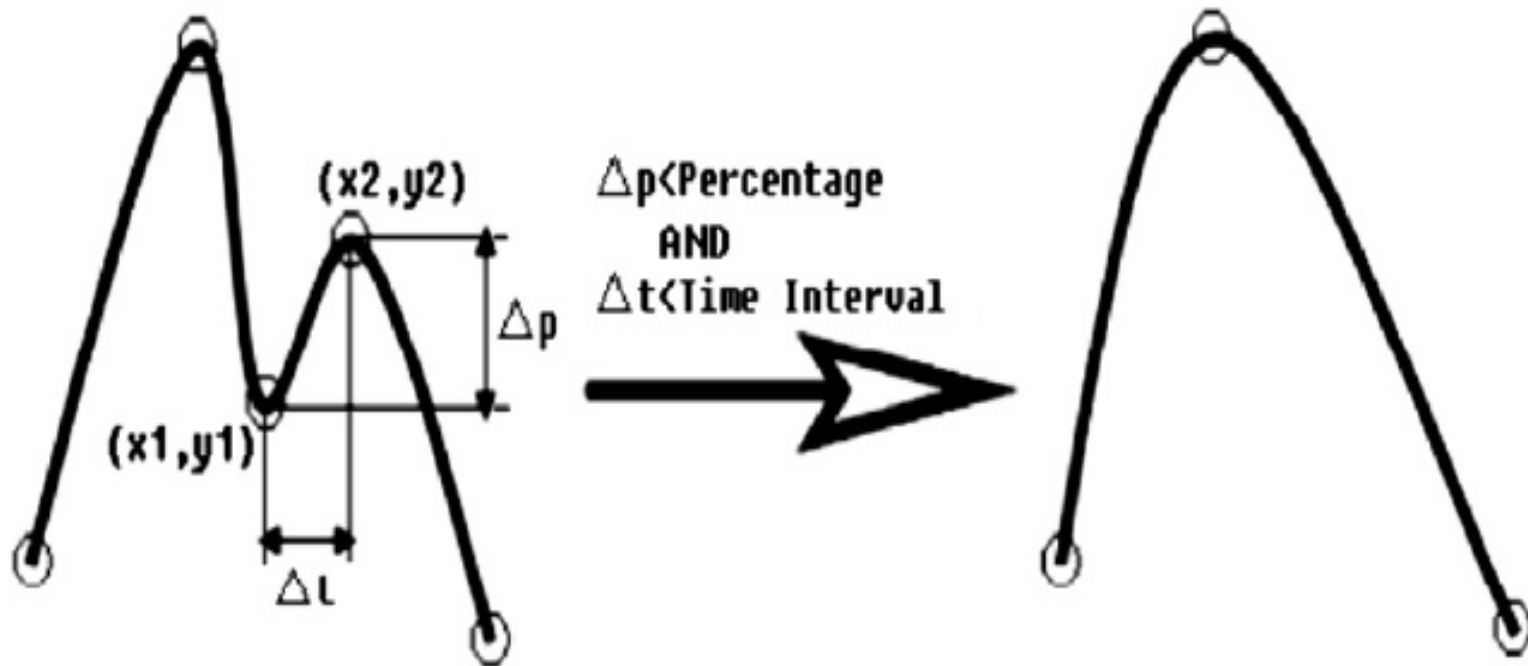


Fig. 4. Criterion of the model.

$$x_{i+1} - x_i < T \text{ and } \frac{|y_{i+1} - y_i|}{(y_i + y_{i+1})/2} < P$$

Ex. : Trader trades once a week(5 business days) and regards a 5% gain or loss as significant, then uses $T = 5$, $P = 0.05$ to recognize turning point.

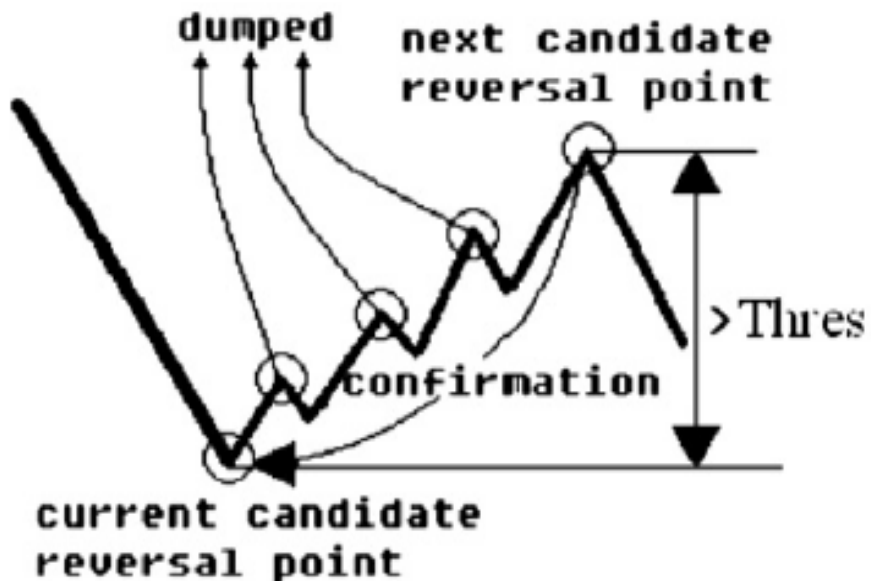
Positive and negative instances selections

Positive instance : critical point that exceeds specified threshold

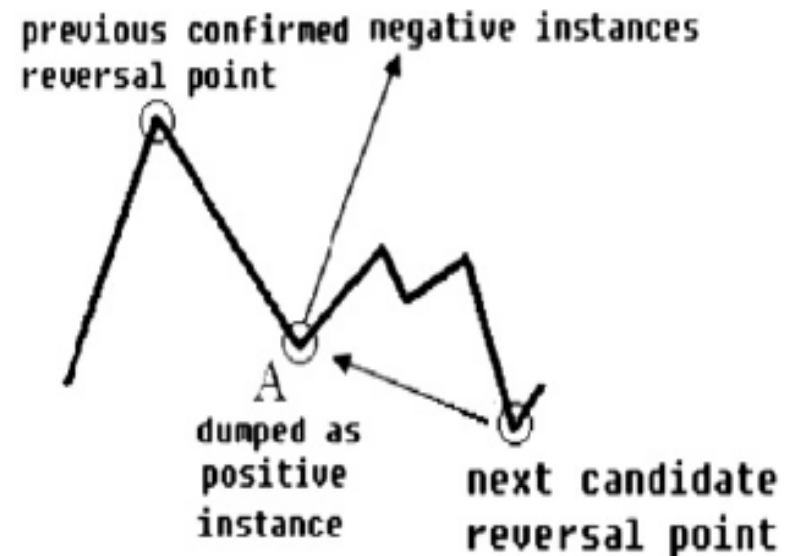
Negative instance : conversely

disadvantage : too many negative instances. How to solve it?

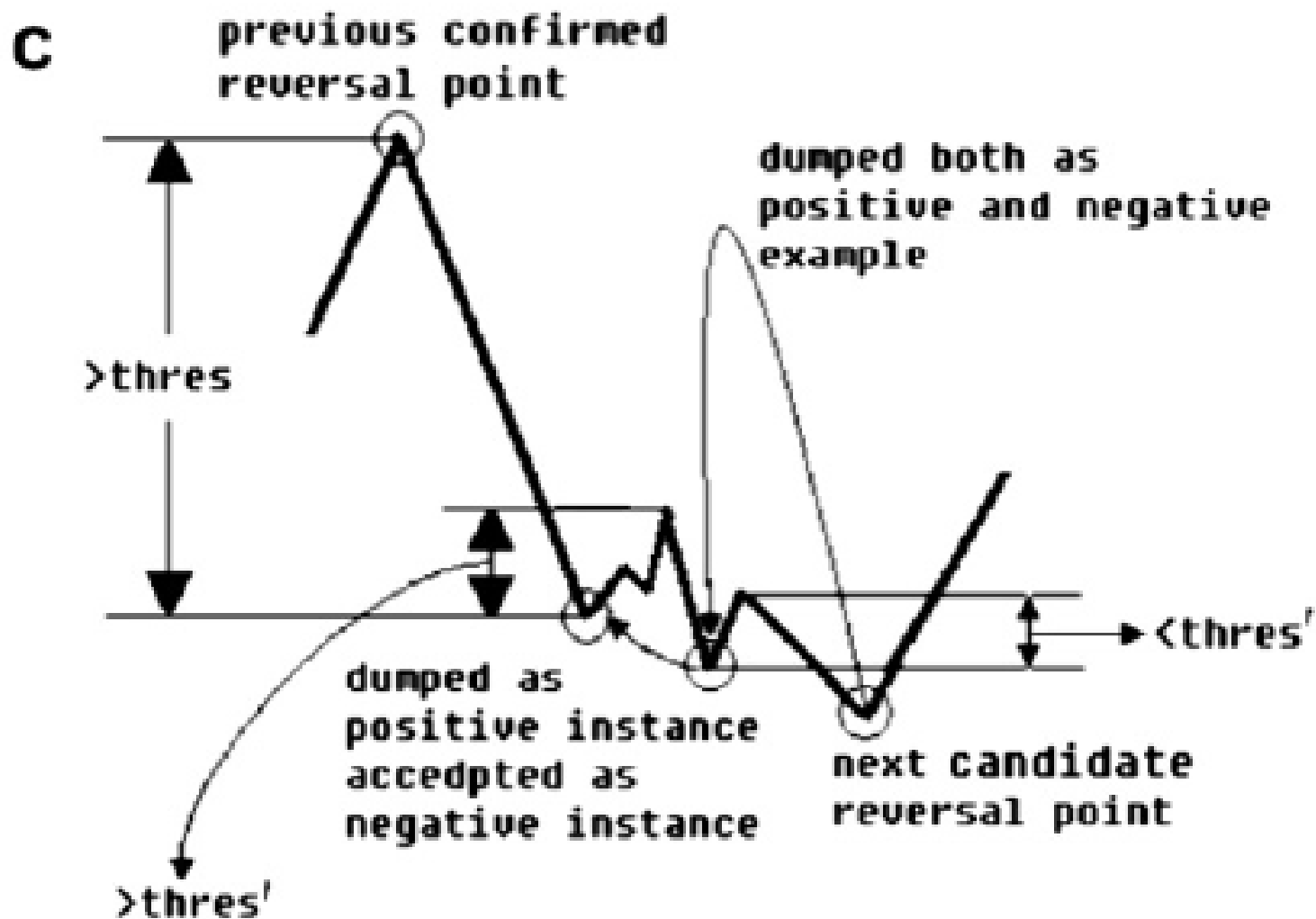
a Neither positive nor negative instances



b

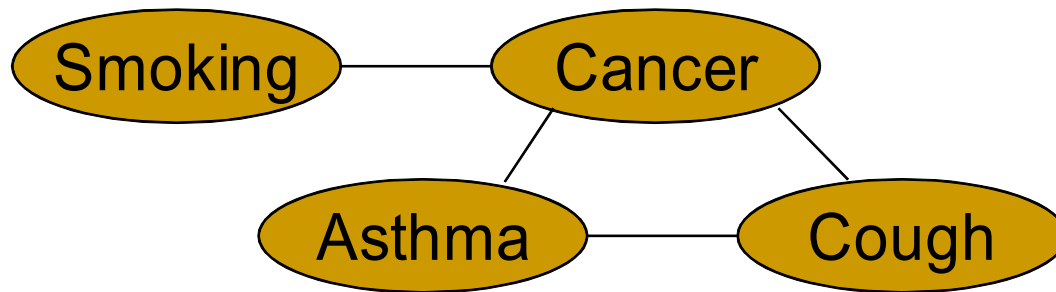


Confirm this reversal until next candidate point exceeding threshold.



Markov Network (Markov Random Field)

- A model of the (full) **joint probability distribution** of a set X of random variables having the **Markov property**.
- Random variable + Undirected graph structure



- Potential functions defined over cliques

$$P(x) = \frac{1}{Z} \prod_c \Phi_c(x_c)$$

$$Z = \sum_x \prod_c \Phi_c(x_c)$$

Smoking	Cancer	$\Phi(S,C)$
False	False	4.5
False	True	4.5
True	False	2.7
True	True	4.5

- Log-linear model:

$$P(x) = \frac{1}{Z} \exp \left(\sum_i w_i f_i(x) \right)$$

Weight of Feature i Feature i

$$f_1(\text{Smoking}, \text{Cancer}) = \begin{cases} 1 & \text{if } \neg \text{Smoking} \vee \text{Cancer} \\ 0 & \text{otherwise} \end{cases}$$

$$w_1 = 1.5 (\because e^{1.5} \approx 4.5)$$

- To maximize the probability is to maximize:

$$\sum_C \log \Phi_i(x_i) = \sum_i w_i f_i(x)$$

- Our model defines $f_i(x)$ in logic (i.e. $f_i(x)$ belongs to 0 or 1)
- Connections between nodes as propositional rules $L_{i1} \rightarrow L_{i2}$ and each rule corresponds to one clique.

Inference

- Inference in Markov networks equal to answer arbitrary queries of the form:

$$P(V_1|V_2) = \frac{P(V_1 \wedge V_2)}{P(V_2)} = \frac{\sum_{x \in \mathcal{X}_{V_1} \cap \mathcal{X}_{V_2}} P(X = x)}{\sum_{x \in \mathcal{X}_{V_2}} P(X = x)}$$

Where $\mathcal{X}_{V_i}$ is the set of truth value.

- In the problem, the conditional probabilities required to calculate take only one form (however computing the equation directly is **intractable**).

$$P(Y = y | X' = \text{TRUE}) = \frac{\sum_{x \in \mathcal{V}_{X' = \text{TRUE}} \cap \mathcal{V}_{Y=y}} P(Y = y, X = x)}{\sum_{x \in \mathcal{V}_{X' = \text{TRUE}} \cap \mathcal{V}_{Y=\text{TRUE}}} P(Y = \text{TRUE}, X = x) + \sum_{x \in \mathcal{V}_{X' = \text{TRUE}} \cap \mathcal{V}_{Y=\text{FALSE}}} P(Y = \text{FALSE}, X = x)},$$

where $X' \subseteq X$.

Parameter learning

- Assume any variable without explicit assignment will be considered as **FALSE**.
- The log-likelihood function:

$$\begin{aligned} L(W) &= \log \prod_{j=1} P_{\omega}(X = x_j) \\ &= \sum_{j=1}^n \sum_i \omega_i f_i(x_j) - n \log Z \end{aligned}$$

where W is weight vector $W = (\omega_1, \omega_2, \dots, \omega_{|n|})$, n is the number of all instances.

$$\frac{\partial}{\partial \omega_i} L(W) = \sum_{j=1}^n f_i(x_j) - n \sum_{j=1}^n P_{\omega}(X = x_j) \cdot f_i(x_j)$$

where the sum is over all possible instances, and $P_{\omega}(X = x_j)$ is $P(X = x_j)$ computed using the current weight vector.

- But inference over the model is **intractable**
- Alternative method : **Discriminative training**
- The parameter estimation through **optimizing conditional likelihood function** is more convenient than **optimizing likelihood function according to the structure of our model**.

$$\prod_{j=1}^n P(Y = y_j | X = x_j) \\ = \prod_{j=1}^n \frac{1}{Z_x} \exp \left(\sum_{i \in V_y} \omega_i \cdot f_i(x_j, y_j) \right)$$

where V_Y is the set of cliques that contains at least one variable in Y .

- The gradient of the conditional log-likelihood(CLL) is :

$$\begin{aligned} \frac{\partial}{\partial \omega_i} \log \prod_{j=1}^n P_{\omega}(Y = y_j | X = x_j) \\ = \sum_{j=1}^n \left[f_i(x_j, y_j) - n \cdot \sum_{y' \in V_Y} P_{\omega}(Y = y' | X = x_j) \cdot f_i(x_j, y') \right] \end{aligned}$$

where V_Y is the set of all possible values of Y .

- Only one query and two possible values in our model

$$\begin{aligned}
 & \frac{\partial}{\partial \omega_i} \log \prod_{j=1}^n P_{\omega}(Y = y_j | X = x_j) \\
 &= \sum_{j=1}^n f_i(x_j, y_j) - \sum_{j=1}^n [P_{\omega}(Y = \text{TRUE} | X = x_j) \\
 & \quad \cdot f_i(x_j, y_j = \text{TRUE}) + P_{\omega}(Y = \text{FALSE} | X = x_j) \\
 & \quad \cdot f_i(x_j, y_j = \text{FALSE})],
 \end{aligned}$$

where

$$\begin{aligned}
 & P_{\omega}(Y = \text{TRUE} | X = x_j) \\
 &= \frac{\exp(\sum_i \omega_i f_i(x_j, y_j = \text{TRUE}))}{\exp(\sum_i \omega_i f_i(x_j, y_j = \text{TRUE})) + \exp(\sum_i \omega_i f_i(x_j, y_j = \text{FALSE}))},
 \end{aligned}$$

$$\begin{aligned}
 & P_{\omega}(Y = \text{FALSE} | X = x_j) \\
 &= \frac{\exp(\sum_i \omega_i f_i(x_j, y_j = \text{FALSE}))}{\exp(\sum_i \omega_i f_i(x_j, y_j = \text{TRUE})) + \exp(\sum_i \omega_i f_i(x_j, y_j = \text{FALSE}))}.
 \end{aligned}$$

Experiment

- **Trading rule** : When we confirm current reversal with certain probability, we sell/buy.
- \$1000 initial fund.
- Each stock will be **trained and tested individually**.
- Only dealing with stock with over 3000 points(days)

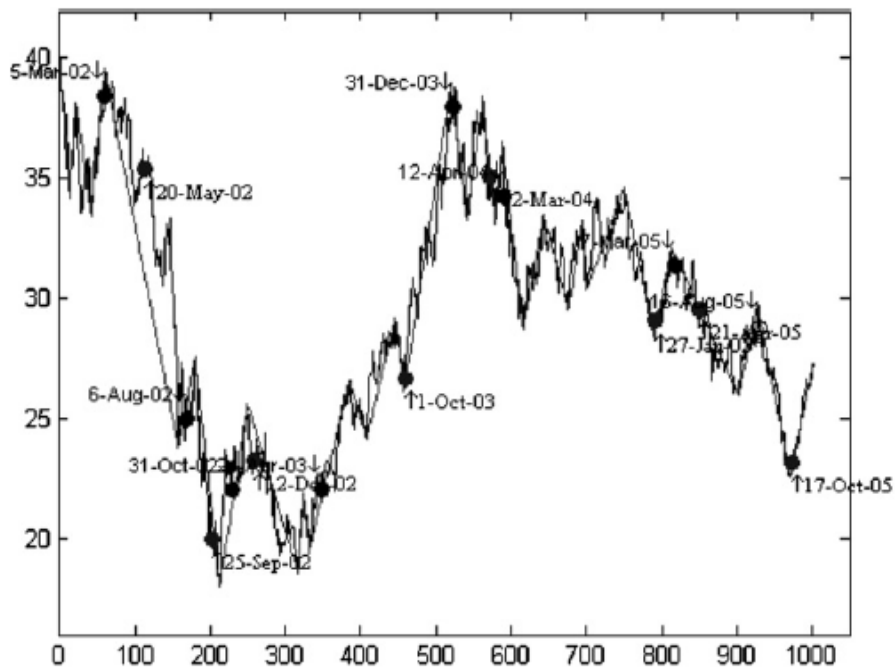


Fig. 6. Trading log (black: sell, grey: buy) of ALCOA INC for 4 years, with probability set to 0.8 and profit 71.2%.

Bear market, Stock loss about 31.6%

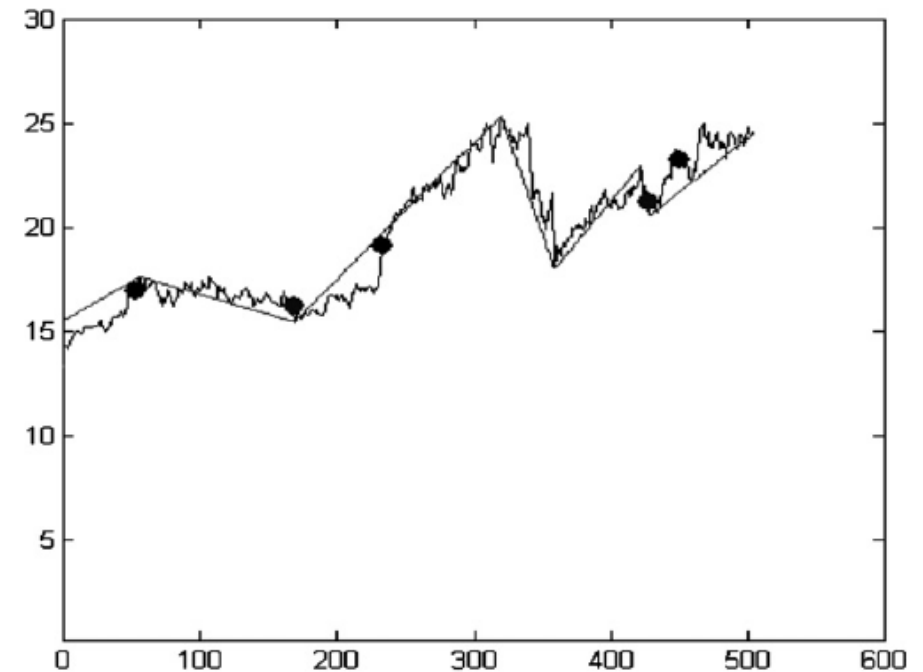


Fig. 7. Trade log of AMERISOURCEBERGEN CP for 2 years, with probability set to 0.8 and profit 79%.

Bull market, stock gains about 61.3%

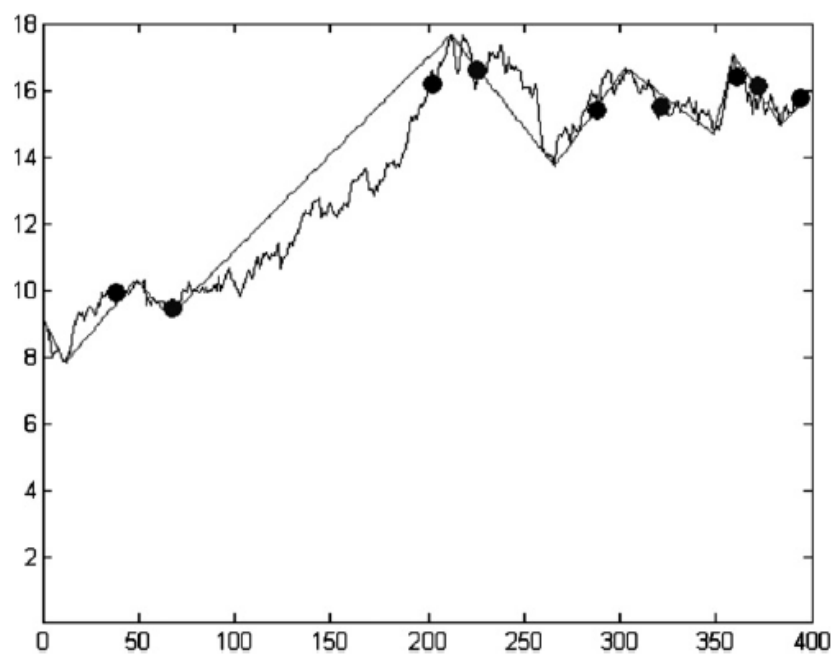


Fig. 8. Trade log of AES CP INC for 2 years, with probability set to 0.8 and profit 104%.

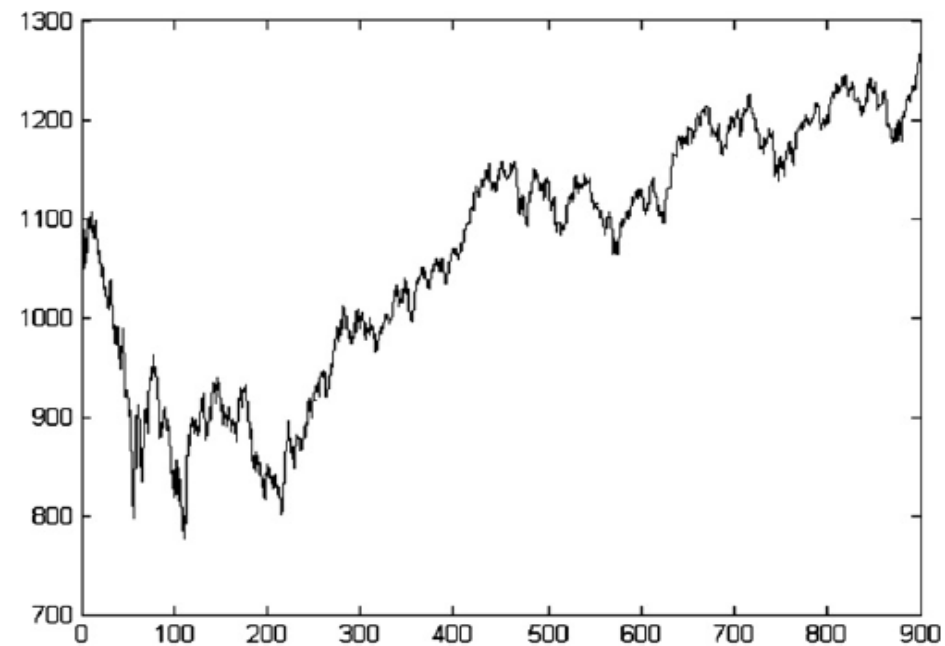


Fig. 9. S & P500c from May 3, 2002 to November 25, 2005 (900 points).

S&P500 index, gains about 18.15%

Table 1. Three periods profit of 10 arbitrary selected stock in S & P500

Symbol	2000–2002 (%)	2002–2004 (%)	2004–2006 (%)
ASN	15.2	64.9	14.5
BHI	71.6	45.4	5.4
CMVT	34.3	35.7	229.5
CVS	9.2	58.4	70.3
HAS	20.8	5.7	201.3
LEN	48.3	37.2	–20.7
LPX	7.0	55.6	12.9
MUR	20.7	32.1	64.4
PGL	–5.9	13.3	33.3
PHM	–26.8	77.8	24.8

Choose 10 stocks randomly with probability 0.8

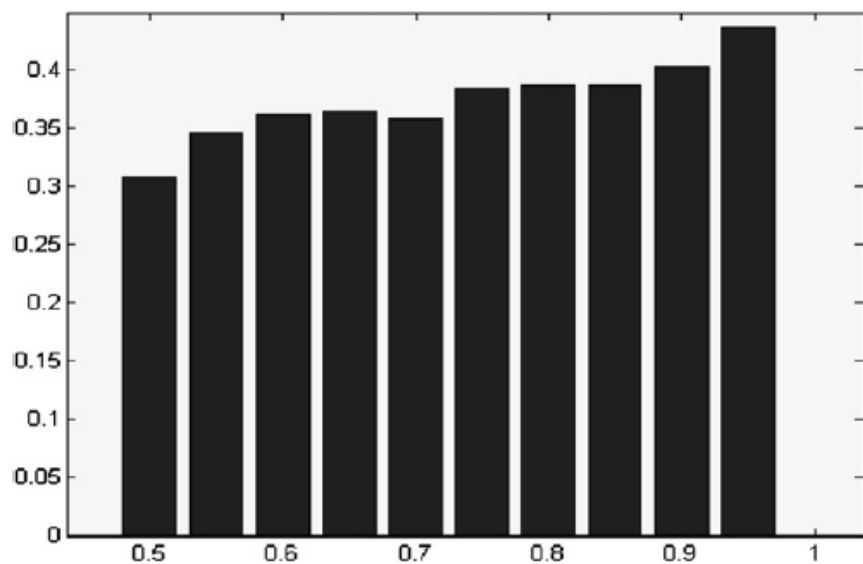


Fig. 10. Average profit rate of S & P500 components on different probabilities.

Trading on 454 stocks and get profits on 416 stocks.

Three specified criteria :

- **Missed opportunity:**
the number of real point without trading operation.
- **False operation:**
the number of loss operations(buy high and sell low) in total operations.(one operation contains a buy and a sell.)
- **profit**

Table 2
Missed opportunities and false operation of IBM

Probabilities	0.5	0.55	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95
Actual opportunity	36	36	36	36	36	36	36	36	36	36
Missed opportunity	4	4	4	4	4	4	4	10	14	18
Missed %	11.1	11.1	11.1	11.1	11.1	11.1	11.1	27.8	38.9	50
Total operation	16	16	16	16	16	16	16	13	11	9
False operation	7	7	5	5	5	3	3	4	3	3
F.O %	43.8	43.8	31.3	31.3	31.3	18.8	18.8	30.8	27.3	33.3
Profit %	24.4	23.3	93.7	92.8	98.9	145.1	160.3	63.0	34.9	53.1

Table 3
Missed opportunities and false operation of microsoft

Probabilities	0.5	0.55	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95
Actual opportunity	48	48	48	48	48	48	48	48	48	48
Missed opportunity	4	4	6	8	8	8	14	24	34	42
Missed %	8.3	8.3	12.5	16.7	16.7	16.7	29.2	50	70.8	87.5
Total operations	22	22	21	20	20	20	17	12	7	3
False operation	9	9	7	5	5	5	5	3	2	1
F.O %	40.9	40.9	33.3	25	25	25	29.4	25	28.6	33
Profit %	-27.1	-23.6	69.5	87.5	75.9	74.3	36.0	21.1	-9.2	-26.9

- Tested from January 3, 2000 to December 30, 2005

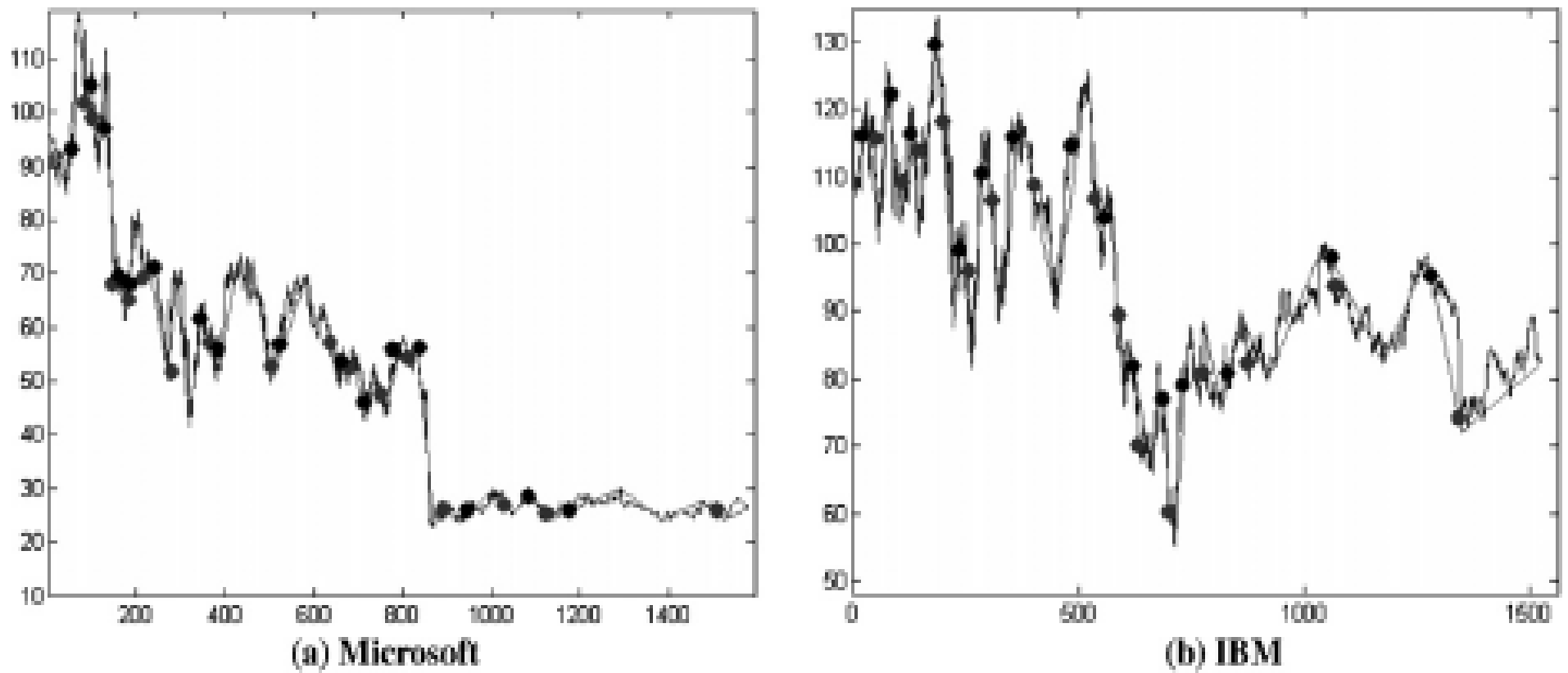


Fig. 11. Trading log (buy: grey sell: black) of IBM and Microsoft.

THANK YOU!!